

Alexandre Champagne-Ruel

NASA Postdoctoral Program Fellow

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Origins of Life • Astrobiology • Complex Systems

RESEARCH PROFILE

Physicist and NASA Postdoctoral Fellow studying why driven chemical systems become complex rather than settle at equilibrium. I use stochastic artificial-chemistry simulations and assembly theory to test how environmental structure and energy flow shape molecular complexity before Darwinian evolution, and I am extending the same questions to biological regulatory networks. The applied aim is an abiotic baseline for agnostic life detection.

ACADEMIC POSITIONS

NASA Postdoctoral Program Fellow

Arizona State University, Biodesign Institute

2025 – Current

Tempe, AZ, USA

EDUCATION

Ph.D. Physics

Université de Montréal

2025

Montréal, QC, Canada

- Thesis: *Spatiality in Prebiotic Evolution: Toward a Physics of the Emergence of Complexity*
- Advisor: Paul Charbonneau

M.Sc. Physics

Université de Montréal

2020

Montréal, QC, Canada

- Thesis: *From Game Theory to Exobiology: The Emergence of Cooperation as a Critical Phenomenon*
- Advisor: Paul Charbonneau

B.Sc. Physics

Université de Montréal

2018

Montréal, QC, Canada

B.Sc. Philosophy

Université de Montréal

2012

Montréal, QC, Canada

GRANTS & AWARDS

Only individual grants >\$5,000 are listed.

NASA Postdoctoral Program Fellowship NASA

Project: Mapping Molecular Complexity for Agnostic Life Detection

146,496 USD

2025

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2024

Globalink Research Award Mitacs Canada

6,000 CAD

2024

Google Cloud Research Grant Google

5,000 USD

2023 – 2026

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2023

J. Armand Bombardier Scholarship <i>Fondation Bombardier</i>	10,000 CAD	2022
Doctoral Scholarship <i>Fonds de Recherche du Québec</i>	70,000 CAD	2022
Excellence Award <i>Université de Montréal</i>	10,000 CAD	2018
Various Travel & Research Grants <i>13 awards</i>	~27,000 CAD	2011 – 2025

PUBLICATIONS

Invited Articles

- A. **Champagne-Ruel** et al., “Complex-systems approaches to astrobiology”, (2027), (Invited Review for Nature Reviews Physics; corresponding author, 18 authors; in preparation).

Accepted Articles

- A. **Champagne-Ruel**, C. P. Kempes, and C. Mathis, “Spatial Patterning and Selection: How the Environment Shapes Molecular Complexity”, [10.48550/arXiv.2509.04547](https://doi.org/10.48550/arXiv.2509.04547) (2026), (in press, PNAS Nexus).

Published Articles

- S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origin of life: An integrated review. Part 1–Experimental methods and data repositories”, *Cell Reports Physical Science* **7**, 103212 (2026).
- S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origins of life: An integrated review. Part 2: Theoretical methods and emerging trends”, *Cell Reports Physical Science* **7**, 103211 (2026).
- S. Rajamani, C. Mathis, S. Asche, D. Boulesteix, C. Blanco, O. Markovitch, A. **Champagne-Ruel**, et al., “Better origin(s) of life research through collaboration across disciplines”, *Cell Reports Physical Science* **7**, 103220 (2026).
- A. **Champagne-Ruel**, S. Zakaib-Bernier, and P. Charbonneau, “Diffusion and pattern formation in spatial games”, *Physical Review E* **110**, 014301 (2024).
- A. **Champagne-Ruel** and P. Charbonneau, “A Mutation Threshold for Cooperative Takeover”, *Life* **12**, 254 (2022).
- S. Gelin, A. **Champagne-Ruel**, and N. Mousseau, “Enthalpy-entropy compensation of atomic diffusion originates from softening of low frequency phonons”, *Nature Communications* **11**, 3977 (2020).

PRESENTATIONS

Invited Talks

Spatiality in Prebiotic Evolution <i>Queens College, City University of New York</i>	2026 <i>New York, USA</i>
Physics of Complexity and Agnostic Life Detection <i>Trottier Institute for Research on Exoplanets</i>	2025 <i>Montréal, QC, Canada</i>
Spatial Patterning and Selection <i>Max Planck Institute for Terrestrial Microbiology</i>	2025 <i>Marburg, Germany</i>
From Emergent Complexity to Reliable Life Detection <i>Arizona State University</i>	2024 <i>Tempe, AZ, USA</i>

Coopération, émergence et transitions

Université du Québec à Montréal

2022

Montréal, QC, Canada

Contributed Talks

How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection

2026

Interdisciplinary Origin of Life Meeting (IOoL)

Tokyo, Japan

Network Topology as a Tunable Parameter for Emergent Complexity

2026

Conference on Complex Systems (CCS)

Binghamton, NY, USA

From Emergent Complexity to Reliable Life Detection

2025

Biennial European Astrobiology Conference (BEACON)

Reykjavik, Iceland

Diffusion: An Overlooked Driver of Prebiotic Complexity

2024

Astrobiology Science Conference (AbSciCon)

Providence, RI, USA

Théorie de l'information et origine de la vie

2023

90e Congrès de l'ACFAS

Montréal, QC, Canada

A Mutation Threshold for Cooperative Takeover

2022

Astrobiology Science Conference (AbSciCon)

Atlanta, GA, USA

Cooperation: An Emergent Universal Feature at the Dawn of Life

2022

Interdisciplinary Origin of Life Meeting for Early Career Researchers

Montréal, QC, Canada

Mutation Favors the Emergence of Cooperation

2021

Life and Space Poland

Virtual/Szczecin, Poland

La criticité dans un système évolutif artificiel

2019

Centre de recherche en astrophysique du Québec (CRAQ)

Orford, QC, Canada

Posters

How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection

2026

Astrobiology Science Conference (AbSciCon)

Madison, WI, USA

Mapping Molecular Complexity for Agnostic Life Detection

2025

NASA Postdoctoral Program Virtual Symposium

Virtual

L'émergence de la coopération via l'évolution de réseaux informationnels

2023

90e Congrès de l'ACFAS

Montréal, QC, Canada

Diffusion, structures spatiales et origine de la vie

2023

90e Congrès de l'ACFAS

Montréal, QC, Canada

OoLEN - The Origin of Life Early-career Network: Building the Community Needed to Solve the Problem

2022

AbSciCon

Atlanta, GA, USA

A Mutation Threshold for Cooperative Takeover

2022

Gordon Research Seminar: Challenging Paradigms in Prebiotic Chemistry

Oxnard, CA, USA

A Mutation Threshold for Cooperative Takeover

2022

Gordon Research Conference: Environments for the Origins of Life and Habitability

Oxnard, CA, USA

Les mutations favorisent la coopération en contexte évolutif

2021

Centenaire, Département de Physique, Université de Montréal

Montréal, QC, Canada

Mutation Favors the Emergence of Cooperative Behavior <i>Molecular Origins of Life Munich</i>	2021 <i>Virtual/Munich, Germany</i>
Mutations Promote Cooperation in an Evolutionary Setting <i>XIXth ISSOL Conference</i>	2021 <i>Virtual</i>
Self-Organized Criticality : A Prelude to Avalanche Models of Solar Flares <i>Space Climate 7 Symposium</i>	2019 <i>Orford, QC, Canada</i>

SCHOOLS & SELECTIVE WORKSHOPS

Assembly Theory for Folded Matter <i>Santa Fe Institute</i>	2025 <i>Santa Fe, NM, USA</i>
Information Driven States of Matter <i>University of Rochester</i>	2024 <i>Rochester, NY, USA</i>

TEACHING EXPERIENCE

Guest Lecturer – CAS 201: Scientific Programming with Python <i>Arizona State University</i>	Fall 2026 <i>Tempe, AZ, USA</i>
Mentor – Biocomputing Scholars program <i>Arizona State University</i>	2025 – 2026 <i>Tempe, AZ, USA</i>
Guiding 3 undergraduate students in developing computational chemistry simulations.	
Undergraduate Internship Supervision <i>Université de Montréal</i>	2022 <i>Montréal, QC, Canada</i>
Supervised a summer research project on diffusion; resulted in journal publication and poster presentation.	
Teaching Assistant – Introduction to Astrobiology <i>Université de Montréal</i>	2021 – 2022 <i>Montréal, QC, Canada</i>
Tutoring – Undergraduate Level <i>Université de Montréal</i>	2018 – 2022 <i>Montréal, QC, Canada</i>

SERVICE & OUTREACH

Scientific Leadership

Positions

Member of the Executive Board <i>Origin of Life Early-career Network (OoLEN)</i>	2022 – 2026
Previously head of the virtual-meetings workgroup; organizer of the OoLEN Public Seminars lecture series (2023–24).	
Founder and Convener <i>cs-astrobio, international journal club and discussion group on complex-systems methods in astrobiology</i>	2026 – Present
Origin of Life Digest (link) <i>Editor</i>	2021 – Present

Conference Organization

Interdisciplinary Origin of Life Meeting

Organizing Committee

2026

Tokyo, Japan

Frontiers in Astrobiology and Origins of Life Conference

Organizing Committee

2025

Reykjavik, Iceland

Origine de la vie : de l'astrophysique à la philosophie (ACFAS congress)

Lead Organizer

2023

Montréal, QC, Canada

Interdisciplinary Origin of Life Meeting for Early Career Researchers

Lead Organizer

2022

Montréal, QC, Canada

Space Climate 7

Local Organizing Committee

2019

Orford, QC, Canada

Session Convener

AbSciCon

Assembly Theory Across Scales: From Molecules to Planetary Systems

2026

Madison, WI, USA

AbSciCon

Exploring Self-Assembly and Self-Organization in Prebiotic Complexity

2026

Madison, WI, USA

Media & Public Engagement

É. Beaudoin-Paul, "Entrevue avec Alexandre Champagne-Ruel, récipiendaire de la prestigieuse bourse en astrobiologie de la NASA", [Quartier Libre](#) (2025).

A. Riopel, "Comment reconnaître la vie sur d'autres planètes", [Le Devoir](#) (2023).

C. Tamura, "Cooperation and the Origin of Life", Quantum Photonics Clubhouse Podcast (2022).

Reviewing Activities

- External Reviewer, NASA Solar System Science Program (2026)
- Journal of the Royal Society Open Science
- Journal of the Royal Society Interface
- Trends in Biochemical Sciences

Memberships

- International Society for Artificial Life
- Center for Research in Astrophysics of Québec
- Canadian Association of Physicists
- Canadian Astronomical Society
- Origin of Life Early-career Network
- International Society for the Study of the Origin of Life
- Complex Systems Society
- Scientific Society for Astrobiology (Founding Member)

TECHNICAL SKILLS

- **Languages:** Fluent in Spoken/Written French, English
- **Programming:** Python, C++, Fortran, Julia, R, LaTeX, MATLAB, x86 assembly, Bash
- **Modeling:** Agent-based, Evolutionary Algorithms, Machine Learning, Network Theory, Game Theory
- **Development Workflow:** Linux, High Performance Computing (HPC)/SLURM, Git, Jekyll, Hugo, HTML5/CSS3